

Thesis Defense



TYSON TEST UNDERGRADUATE
HONORS THESIS



Does Demographic Aging & Financial Globalization Moderate Crowding Out? Federal Deficit Financing and U.S. Long-Term Real Rates.

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Key Takeaways (preview of results)

Modern era (1991-2023): deficits are associated with higher real long-term rates, but the deficit sensitivity declines as the population ages (negative Def/GDP * OADR)

Global conditions matter in the modern era: adding open-economy controls (e.g. German real rate proxy) materially improves fit and changes the deficit association

Earlier era (1960-1990): the deficit-aging interaction differs in sign from the modern era (deficit sensitivity increases with aging)

Research Question & Empirical Approach



Question: How do U.S. Federal Government deficits relate to long-term real rates?



Conditioning: Does population aging change this relationship?



Model Design: Annual U.S. 1960-2023, compare 1960-1990 vs. 1991-2023.



Model: OLS, featuring an interaction term (Deficit/GDP * OADR).



Modern era focus: Inclusion of open-economy controls (global financing conditions).

Why Deficits & Aging Matter for Real Long- Run Rates

Macroeconomic Theory: Crowding-Out Hypothesis, higher government borrowing can raise real long-run rates (holding all else constant).

Aging may shift saving/investment balance and raise safe-asset demand → ambiguous net effect on real rates

Modern Era: deeper financial globalization → global real rate conditions may influence domestic long-term rate movements

Empirical Goal: document conditional associations

Literature Review I: deficits, saving, and long-run real rates

- **Ricardian Equivalence is Empirically Weak:** borrowing is not fully offset by higher private saving; thus, national saving rates are reduced
- **Debt/deficit sensitivity of long-term rates:** empirical estimates generally show a positive (but modest) increase in rates from higher levels of deficits & debt
- **Implication for this thesis:** crowding out is plausible, though magnitude of the effect is uncertain
- **Selected References:** Modigliani & Sterling (1986), Gale & Orszag (2004), and Engen & Hubbard (2004)



Literature Review II: Demography & The Open Economy



- **Demography and secular real-rate decline:** aging-related life-cycle saving patterns help explain trend declines in real rates
 - **Demographics improve bond-market fit** when embedded in term-structure models (slow-moving, predictable shifts in bond demand)
 - **Open-economy/financial globalization:** U.S. shocks transmit globally; cross-border capital flows can be quantitatively important for long-term rates especially in the modern era
 - **Implication for this thesis:** deficit effects may be conditioned by aging and global financing conditions, motivating an interaction and modern era model
 - **Selected references:** Carvalho, Ferrero & Nechio (2017); Favero & Gozluklu (2015); Miranda-Agrippino & Rey (2020).
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Data Sources & Variable Construction

Variable Construction

- Real rate (DV): nominal 10-yr. Treasury yield minus CPI inflation (ex post)
$$r_t^{US} = GS10_t - \pi_t$$
- OADR: (65+ per 100 aged 15-64)
$$OADR_t = 100 * \frac{Pop_{65+t}}{Pop_{15-64_t}}$$
- Key Interaction term: marginal impact of the deficit on real rates at different OADR Levels
$$Def_t * OADR_t$$
- Deficit term coded so positive = deficit (for easier reading of results)

Data Sources

- CPI Inflation: BLS CPI-U (annual; constructed from monthly CPI data)
- Nominal 10-year Treasury Yield (GS10): FRED
- Unified Deficit/Surplus: FRED
- FYFSGDA188S
- OADR: World Bank SP.POP.DPND.OL
- Modern era controls: German real
- 10yr. rate, net capital inflows, personal savings rate

Note: all variables are in % or % of GDP unless otherwise stated

Regression Models Employed

- Regression model (estimated separately for early and modern eras):

$$r_t = \beta_0 + \beta_1 Def_t + \beta_2 OADR_t + \beta_3 (Def_t \times OADR_t) + \gamma' X_t + \varepsilon_t$$

- Baseline model excludes controls
- Key object of interest is the interaction between deficits and demography:

$$\frac{\partial r_t}{\partial Def_t} = \beta_1 + \beta_3 (OADR_t)$$

Note: OLS, robust (Huber-White) standard errors.



Evidence: Split Sample & Interpretation

Modern era: Global Rate Conditions (Preferred Spec)

Interpretations

- Adding open economy controls materially changes the modern era deficit association
- German real rates (global proxy) is strongly related to U.S. real 10-year rates
- Deficit coefficient is positive, but Def/GDP*OADR is negative
- Marginal effect of the deficit is less

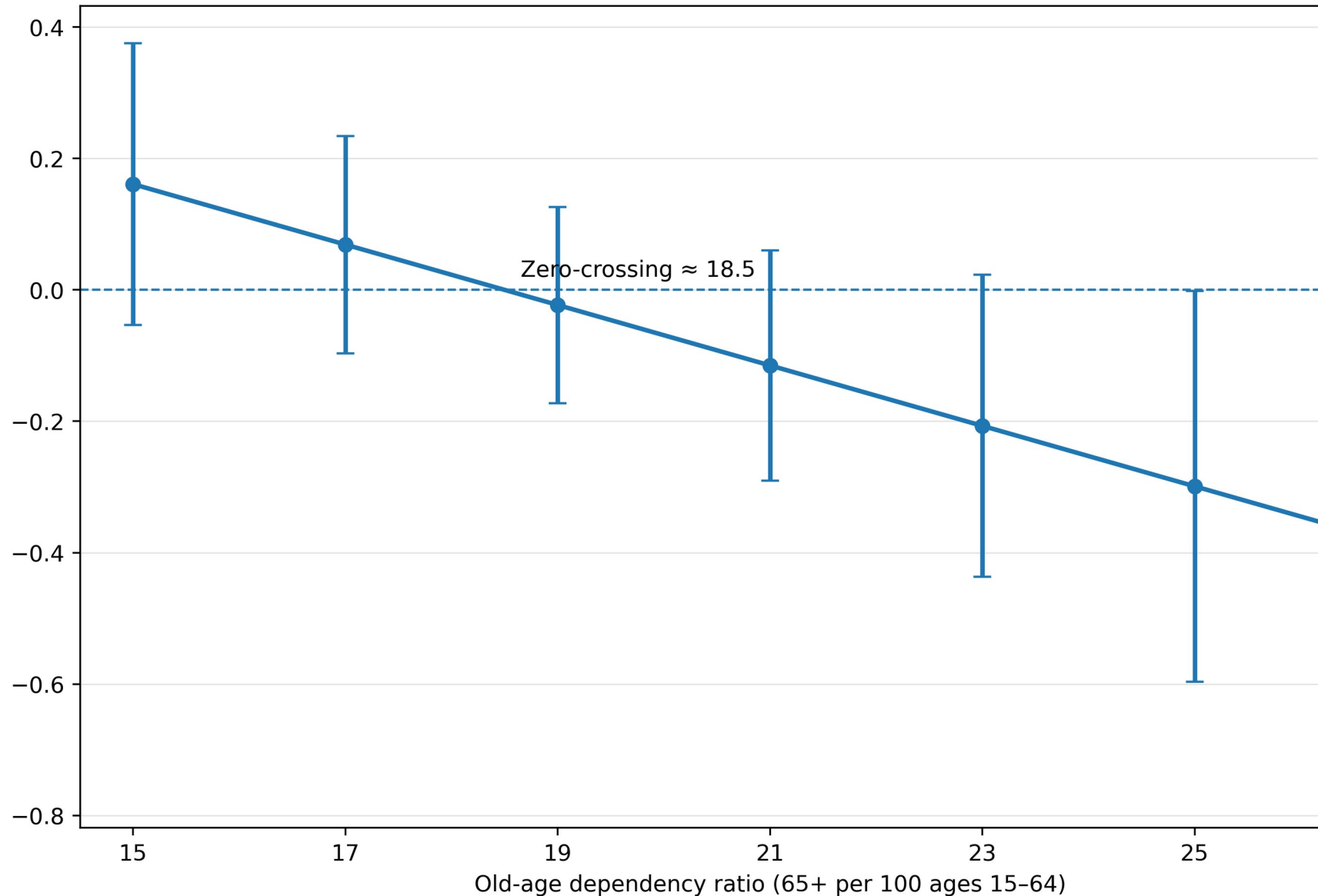
Key Coefficients

- German real 10-year rates: 0.945* (0.218)
- Deficit (% of GDP): 0.851 (0.401)
- Def/%GDP * OADR: -0.046 (0.021)
- N = 33
- Robust (Huber–White) SEs in parentheses.
- R²: baseline 0.538 → preferred 0.892 (post-1990).

Marginal effect of deficits across aging (Modern era model)

- Points show estimated marginal effects; vertical bars denote 95% confidence intervals (robust Huber-White SEs).

- Y-axis: change in real 10-yr. rate (pp) per +1 pp deficit/GDP



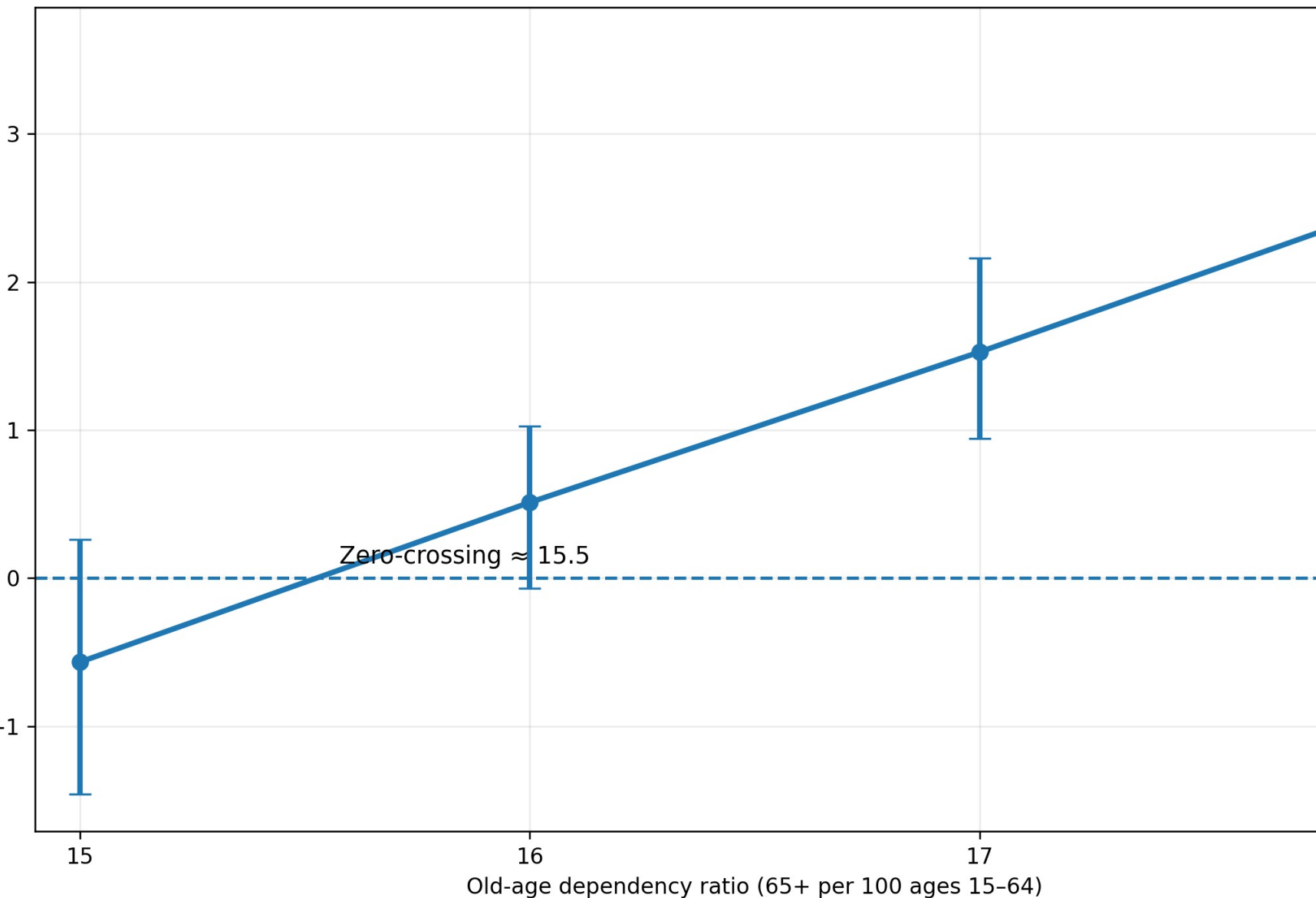
Earlier era baseline: deficit-aging interaction differs from the modern era

Pre-1991 Baseline (1960-1990)	Deficit (% of GDP): -18.218* (4.379)
OADR: -2.775* (0.737)	Deficit * OADR: 1.172* (0.267)
N=30 R ² = 0.598	Zero crossing OADR=15.6

Robust (Huber-White) SE reported here

Interpretation (conditional on interaction)

- Interaction model $\Rightarrow \frac{\partial r_t}{\partial Def_t} = \beta_1 + \beta_3(Def_t \times OADR_t)$
- Main effects are evaluated at OADR=0; interpret the effect within the observed OADR range (15-18)
- Pre-1991: $\beta_3 > 0 \Rightarrow$ deficit sensitivity increases as OADR rises
- Post-1990 preferred: $\beta_3 < 0 \Rightarrow$ deficit sensitivity declines as OADR rises



Marginal effect of deficits across aging (pre-1991 model)

- Points show estimated marginal effects; vertical bars denote 95% confidence intervals (robust Huber-White SEs)
- Y-axis: change in real 10-yr. rate (pp) per +1 pp deficit/GDP
- Zero crossing ≈ 15.6

Cross-era comparison: deficit sensitivity v. aging

EARLY ERA (1960-1990)

- Deficit $\times$ OADR interaction: positive $\beta_{3(Def_t \times OADR_t)} > 0$
- As OADR rises: deficit sensitivity increases
- Zero crossing: OADR ≈ 15.6

MODERN ERA (1991-2023)

- Deficit $\times$ OADR interaction: negative $\beta_{3(Def_t \times OADR_t)} < 0$
- As OADR rises: deficit sensitivity decreases
- Zero crossing: OADR ≈ 18.5

Takeaway: The deficit–real rate relationship appears regime-dependent and conditional on demographics.

Why does deficit sensitivity differ across eras?

- **Financing (quantities):** post-1990 capital market openness and foreign inflows can reduce the need for domestic rate adjustments when deficits rise
- **Global rate level (prices):** American long-run real rates co-move with global real-rate conditions, shifting the baseline around which fiscal effects operate
- **Demographic channel:** higher OADR changes saving/portfolio demand for safe long-duration assets, potentially muting crowding-out at higher OADR
- **Interpretation:** evidence is consistent with **conditional crowding-out**, the deficit-real rate relationship depends on the global environment and demographics



Limitations



Endogeneity (reverse-causality): deficits and real rates can move together because higher interest rates raise net interest outlays, widening the deficit



Macroeconomic shocks can affect both real rates and deficit spending simultaneously



Omitted Variables: long-term slow-moving forces such as total factor productivity growth and different bank regulatory environments



Sample size & regime split: splitting the data into 1960-1990 and 1991-2023 sub-sets reduces sample size for each model



Measurement: real-rate proxy is ex post (GS10 – CPI) and annual aggregation can differ from expectations-based real rates

Conclusion

Deficit-real rate relationship is conditional and regime dependent.

Policymakers should not assume a constant crowding-out effect: the rate impacts of deficits is dependent on demographics and financial globalization

Risk-management framing: if the regime shifts (e.g. weaker foreign capital inflows), deficit sensitivity could return to previously recorded levels, raising long-term financing costs

Appendix

- **A1.** VIF diagnostics
- **A2.** Kitchen-sink regression (selected)
- **A3.** Pairwise correlations
- **A4.** Descriptive statistics (N=64)
- **A5.** Model + marginal-effects interpretation

Appendix A1: VIF diagnostics (Table A1)

Panel A. Core-controls spec (Table 2, col. 4)		Mean VIF = 33.736
Variable	VIF	Tolerance
Deficit (%GDP)	106.777	0.009
OADR	10.078	0.099
Deficit×OADR	107.610	0.009
German real 10y rate	3.528	0.283
Unemployment rate (%)	3.715	0.269
Personal saving rate (%)	2.386	0.419
Federal receipts (% of GDP)	2.054	0.487

Panel B. Saturated (“kitchen-sink”) spec (Table A2)		Mean VIF = 190.220
Variable	VIF	Tolerance
Deficit (%GDP)	1,445.217	0.001
OADR	113.851	0.009
Deficit×OADR	1,449.401	0.001
Federal receipts (% of GDP)	13.852	0.072
German real 10y rate	18.409	0.054
Net equity inflow (% GDP)	6.632	0.151
Net debt inflow (% GDP)	13.147	0.076
Open market operations (% GDP)	7.804	0.128
U.S. investment (% of GDP)	8.568	0.117
U.S. public debt (% of GDP)	76.709	0.013
Discretionary outlays (% of GDP)	85.639	0.012
Net interest (% of GDP)	23.636	0.042

Panel C. Post-1990 marginal-effects spec (Table 3, col. 4)		Mean VIF = 48.504
Variable	VIF	Tolerance
Deficit (%GDP)	121.541	0.008
OADR	5.910	0.169
Deficit×OADR	153.031	0.007
Term spread (10y–3m)	2.408	0.415
Natural unemp. rate (%)	4.840	0.207
Personal saving rate (%)	3.293	0.304

Notes: High VIFs are expected for an interaction term and its components (mechanical correlation), and in saturated specifications where many variables trend together. Emphasize implied marginal effects when interpreting interaction models.

Saturated (“kitchen-sink”) regression (sensitivity check)

Selected coefficients (robust SEs in parentheses)			N=41 R²=0.953	
Variable	Coef.	Robust SE		
Deficit (%GDP)	-0.041	(1.287)		
OADR	1.542***	(0.450)		
Deficit×OADR	-0.024	(0.055)		
German real 10y	0.564***	(0.188)		
U.S. investment (%GDP)	0.654***	(0.222)		
U.S. public debt (%GDP)	-0.113**	(0.049)		
Unemployment rate (%)	1.226***	(0.239)		
Constant	-29.810**	(13.400)		

Other controls included (not shown): receipts; equity/debt inflows; open market ops; discretionary outlays; net interest; saving; term spread; Baa spread; market cap; equity risk premium; natural unemployment rate. Full list in Appendix Table A2.

Correlation structure (core variables)

With U.S. real 10y rate		Among key regressors	
German real 10y	0.880	OADR vs German real 10y	-0.877
OADR	-0.734	Deficit vs Unemployment	0.583
Unemployment rate	0.333	Deficit vs Federal receipts	-0.581
Saving rate	0.325	Deficit vs Saving rate	0.576
Deficit (%GDP)	-0.296	Unemployment vs Receipts	-0.572

Takeaway: Strong co-movement (e.g., OADR with German real rates) supports caution when interpreting individual coefficients; it also motivates focusing on marginal effects in the interaction model.

Descriptive statistics (core-controls variables, 1960-2023)

Mean, SD, and range (annual; N=64 for all variables)

Variable	Mean	SD	Min	Max
U.S. real 10y rate	2.045	2.480	-5.048	8.138
Deficit (%GDP)	3.018	2.962	-2.305	14.669
OADR	18.320	2.786	14.743	26.829
DeficitxOADR	59.934	67.599	-42.135	359.746
German real 10y rate	2.619	2.387	-5.362	6.296
Unemployment rate (%)	5.917	1.618	3.490	9.708
Personal saving rate (%)	8.451	3.244	2.250	15.092
Federal receipts (%GDP)	17.852	1.006	15.485	20.173

Notes: Variables correspond to the core-controls specification (Table 2, col. 4).

Model and marginal effects (interaction interpretation)

Estimated model (OLS):

$$r_t = \beta_0 + \beta_1 \text{Deficit}_t + \beta_2 \text{OADR}_t + \beta_3 (\text{Deficit}_t \times \text{OADR}_t) + \gamma'X_t + \varepsilon_t$$

Marginal effect of deficit:

$$\partial r_t / \partial \text{Deficit}_t = \beta_1 + \beta_3 \cdot \text{OADR}_t$$

- Sign of β_3 determines whether deficit sensitivity rises or falls with aging.
- 'Zero-crossing' occurs at $\text{OADR} = -\beta_1/\beta_3$ (within observed OADR range).
- β_1 alone corresponds to $\text{OADR}=0$ (outside the data range), so interpret effects within observed OADR.

Use: backup slide if asked how to interpret interaction coefficients or why the 'main effect' looks large.